

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ELECTRONICS AND INSTRUMENTATION ENGINEERING

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1EI401	Architecture and Protocols of Internet of Things	3	0	0	3	3
22PC1EI402	Analytical Instrumentation	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1EI401	Drives and Control systems for Robotics	3	0	0	3	3
22PE1EI402	Instrumentation for Petrochemical industries					
22PE1EC407	Speech Signal Processing					
22PE1EC421	Wireless Sensor Networks and Protocols					
22PE1EI403	Biomedical Signal Processing					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Nanotechnology Stream	3	0	0	3	3
	Power Electronics Stream					
	Industrial Internet of Things Stream					
	Embedded systems Stream					
	Biomedical Imaging Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2EI401	IoT Applications Laboratory	0	0	2	2	1
22PC2EI402	Analytical Instrumentation Laboratory	0	0	2	2	1
22PW4EI401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ELECTRONICS AND INSTRUMENTATION ENGINEERING

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1EI404	Robot Programming and Intelligent Control System	3	0	0	3	3
22PE1EI405	Instrumentation for Pharmaceutical Industries					
22PE1EI406	Neural Networks and Fuzzy Logic Systems					
22PE1EC402	DSP Processors and Architectures					
22PE1EI407	Tele Medicine					
	PROFESSIONAL ELECTIVE – VI					
22PE1EI408	Micro Electro-Mechanical Systems	3	0	0	3	3
22PE1EI409	Pollution Control in Process Industries					
22PE1EC305	Digital Image Processing					
22PE1EC420	CPLD and FPGA Architectures					
22PE1EI410	Automotive Instrumentation					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4EI402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1EI401) ARCHITECTURE AND PROTOCOLS OF INTERNET OF THINGS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the Internet of Things eco-system
- To impart knowledge of architecting Internet of Things
- To understand the protocols of the Internet of Things at various levels
- To develop skills required for an IoT architect
- To gain knowledge of the Cloud Protocols

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Concisely establish knowledge about the IoT ecosystem and applications

CO-2: Understand the concepts of IoT Architecture Reference model and IoT reference architecture

CO-3: Illustrate various protocols and standards used in various layers of IoT architecture

CO-4: Analyze and apply various IoT Edge to Cloud Protocols

CO-5: Build and demonstrate the project successfully by sensor requirements, coding, emulating and testing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	3	3	1	1	-	-	-	-	-	1	3	1	3
CO-2	3	2	3	3	1	1	-	-	-	-	-	1	2	1	3
CO-3	3	2	3	3	1	1	-	-	-	-	-	1	2	1	3
CO-4	3	1	3	3	1	1	-	-	-	-	-	1	2	1	3
CO-5	3	1	3	3	1	2	-	-	-	-	2	1	2	1	3

UNIT-I:

Introduction - IoT and digitization IoT impact Convergence of Information Technology and Operational Technology

IoT Architecture: The Key Drivers for the IoT Discipline, The Diversity of IoT Data Sources, The Connected Device, IoT ecosystem, IoT Vs machine to machine, IoT Architecture, Role of an architect

UNIT-II:

Protocols for IoT Eco-System: Layered Architecture for IoT, Protocol Architecture of IoT, Categorization of IoT protocols

WPAN Standards: 802.15 standards, Bluetooth5- communication process and topologies, protocol stack, IEEE 802.15.4, Zigbee, Z-wave architecture, topology.

UNIT-III:

IP-Based WPAN and WLAN: Internet Protocol and Transmission Control Protocol-IP role in IoT, 6LoWPAN- topology, Thread- architecture and topology, IEEE 802.11- suite of protocols and comparison, architecture

Long-range Communication Systems and Protocols (WAN): Cellular Connectivity-4G-LTE topology, 5G-VoLTE, architecture and Protocol Stack, LoRa and LoRaWAN.

UNIT-IV:**IoT Edge to Cloud Protocols:**

MQTT: publish-subscribe, architecture, packet structure, communication formats, working example MQTT-SN: architecture and topology, transparent and aggregating gateways, Gateway advertisement and discovery.

Constrained Application Protocol: architecture, Messaging Formats, Usage Example, STOMP, AMQP, and Comparison of Protocols.

UNIT-V:

Programming the Microcontroller for IoT: Working principles of sensors – IoT deployment for Raspberry Pi /Arduino/Equivalent platform – Reading from Sensors, Communication: Connecting microcontroller with mobile devices - Communication through Bluetooth, WiFi and USB

TEXT BOOKS:

1. Internet of Things for Architects: Architecting IoT Solutions by Implementing Sensors, Communication Infrastructure, Edge Computing, Analytics, and Security, Perry Lea, Packt Publishing, 2018
2. Internet of Things: Architectures, Protocols and Standards, Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri, Wiley, 2019

REFERENCES:

1. The Internet of Things: Enabling Technologies, Platforms, and Use Cases, Pethuru Raj and Anupama C. Raman, CRC Press
2. The Internet of Things in the Cloud: A Middleware Perspective, Honbo Zhou, CRC Press, 2012
3. Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Michahelles, Florian, Springer, 2011
4. Networks, Crowds, and Markets: Reasoning About a Highly Connected World, David Easley and Jon Kleinberg, Cambridge University Press, 2010
5. The Internet of Things – Key Applications and Protocols, Olivier Hersent, David Boswarthick, Omar Elloumi, Wiley, 2012

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1EI402) ANALYTICAL INSTRUMENTATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the whole array of modern analytical instruments to implement statical analysis tools
- To familiarize with sample preparation, application, method development, data analysis and interpretation being key elements
- To introduce methods to Interpret data derived from any analytical instrument.
- To familiarize with the basic concepts, principles and terms of chromatography, Spectroscopy and gas analyzers

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Examine the relevant methods and techniques to analyze samples

CO-2: Demonstrate proficiency in using statistical methods for evaluating and interpreting data using spectroscopy, chromatography and gas analyzers

CO-3: Identify tools to apply principles of spectroscopy, chromatography, and gas analyzers

CO-4: Apply analytical techniques for day to day and industrial applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	3	-	1	1	-	1	2	1	-	1	-	2	-
CO-2	1	3	1	-	2	2	-	2	1	2	-	2	-	3	-
CO-3	3	1	2	1	2	3	-	2	2	3	-	2	-	2	-
CO-4	2	1	2	-	1	1	-	2	1	2	-	3	-	2	-

UNIT-I:

Electrochemical Instruments: Basic concepts of Analytical Instrumentation, Electro chemical instruments- pH meter, Conductivity meter, Dissolved oxygen analyzers, sodium analyzers, silica analyzers

UNIT-II:

Absorption Spectroscopy: Concepts of Spectrometry, Beer- Lambert's law, Derivation of Beer Lamberts Law- Numerical exercises associated with the Law. UV, VIS spectrophotometers – single beam and double beam spectrophotometers – Instrumentation associated with the above spectrophotometers – sources and detectors. IR Spectrometers – Sources and detectors, Instrumentation associated with the above spectrophotometers, FTIR. Interpretation and Analysis.

Emission Spectroscopy: Flame emission and Atomic emission spectrophotometers – Sources for Flame Photometers, Online calorific value measurements.

UNIT-III:

Chromatography: Chromatography – types- Basic principles of gas chromatography, liquid chromatography (HPLC) -different types of columns, detectors, recorders and associated equipment for Gas and Liquid Chromatographs and their applications, Interpretation and Analysis.

UNIT-IV:

Nuclear Magnetic Resonance Spectrometry: Instrumentation associated with NMR spectrophotometer – Introduction to mass spectrophotometers, Introduction and Working Principle of Electron Spin Resonance (ESR), Interpretation and Analysis.

Gas Analyzers: Analysis using Thermal conductivity principle, Katharometer – oxygen analyzers using paramagnetic principle, H₂S analyzer system, CO monitors, NO_x analyzers, Industrial analyzer circuits, Pollution Monitoring systems.

UNIT-V:

Thermal Analyzers and Nuclear Radiation Detectors: Differential Scanning Calorimetry (DSC), Derivative Thermo Gravimetric Analyzers (DTGA).

Gas-filled Detectors- GM counters, Scintillation counter, Ionization chamber, Proportional counter, solid state detector.

TEXT BOOKS:

1. Handbook of Analytical Instrumentation, R. S. Khandpur, 3rd Edition, Tata McGraw-Hill
2. Instrumental Method of Analysis, Willard H. H., Merrit L. L., Dean, D. Van Nostrand, 7th Edition, CBS Publishing and Distributors, 1995

REFERENCES:

1. Process Measurement and Analysis, B. G. Liptak, 5th Edition, CRC Press, 2016
2. Principles of Instrumental Analysis, Skoog D. A. and West D. M., 7th Edition, Holt Sounder Publication, 1985
3. Instrument Technology, Jones B. E., 3rd Edition, Butterworth Scientific Publications, 1987

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EI401) DRIVES AND CONTROL SYSTEMS FOR ROBOTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To get acquainted with different robot drive mechanisms
- To understand in detail the working of hydraulic and pneumatic drives used in robotics
- To learn the working principles of various electric drive systems for robotics
- To acquire basic Knowledge of servo systems for robot control

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Categorize various drive systems for robot movement

CO-2: Select the appropriate drive system for a particular application

CO-3: Inspect different electric drives and their applications in robotics

CO-4: Analyze the accurate positioning of the robot end effector by the servo control

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	3	-	-	-	-	-	1	2	2	1	3
CO-2	3	1	1	-	3	-	-	-	-	-	-	2	2	2	2
CO-3	3	1	-	-	1	1	1	-	-	-	-	-	2	2	2
CO-4	3	3	2	1	1	-	-	-	-	-	-	1	1	2	2

UNIT-I:

Introduction: Objectives, motivation, open loop control system, closed loop control system with velocity and position feedback, Types of drive systems. Functions of drive system.

Robot Drive Mechanism: Lead Screws, Ball Screws, Chain & linkage drives, Belt drives, Gear drives, Precision gear boxes, Harmonic drives.

UNIT-II:

Hydraulic Drives: Introduction, Requirements, Hydraulic piston and transfer valve, hydraulic circuit incorporating control amplifier, hydraulic fluid considerations, hydraulic actuators, Rotary and linear actuators. Hydraulic components in robots.

UNIT-III:

Pneumatic Drives: Introduction, Advantages, Pistons - Linear pistons, Rotary pistons, Motors-Flapper motor, Geared motor, Components used in pneumatic control. Pneumatic proportional controller, pneumatically controlled prismatic joint.

UNIT-IV:

Electric Drives: Introduction, Types, DC electric motor, AC electric motor, stepper motors, half-step mode operation, micro-step mode. Types of stepper motors, including direct drive actuators.

UNIT-V:

Servo Mechanism for Robot: Mathematical modelling of robot servos, error responses and steady state errors in robot servos, feedback and feedforward compensations, hydraulic position servo, computer-controlled servo system for robot applications, selection of the robot drive systems.

TEXT BOOKS:

1. Fundamentals of Robotics Analysis and Control, Robert J. Schilling, PHI Learning, 2009
2. Robotics Engineering - An Integrated Approach, Richard D. Klafter, Thomas A., Chri Elewski, Michael Negin, PHI Learning, 2009

REFERENCES:

1. Industrial Robotics, Technology Programming and Applications, Mikell P. Groover & Nicholas G. Odrey, Mitchel Weiss, Roger N. Nagel, Ashish Dutta, Tata McGraw-Hill Education, 2017
2. Industrial Robotics, Bernard Hodges, 2nd Edition, Jaico Publishing House, 1993
3. Foundations of Robotics Analysis and Control, Tsuneo Yohikwa, MIT Press, 2003
4. Introduction to Robotics Mechanics and Control, John J. Craig, 3rd Edition, Pearson, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EI402) INSTRUMENTATION FOR PETROCHEMICAL INDUSTRIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To expose to various petroleum production processes
- To impart knowledge on various processes involved in petroleum refinery
- To provide knowledge on specific measurement techniques practiced, control systems and automation involved in the petrochemical industry

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand various petrochemical processes and important parameters to be monitored and controlled

CO-2: Identify various instruments that are used to control the petrochemical process

CO-3: Have an understanding of the production routes for important petrochemicals

CO-4: Define automation and safety standards of a petrochemical industry

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	-	-	1	1	1	2	-	-	1	-	-	1	2	2
CO-2	1	1	-	2	1	2	2	-	1	1	1	1	1	2	1
CO-3	1	1	1	-	1	2	2	-	1	1	1	1	1	2	1
CO-4	1	1	1	-	-	2	1	-	2	2	2	2	1	3	2

UNIT-I:

Introduction to Petrochemical Products: Brief survey of petroleum formation, petroleum exploration, Petroleum production, Petroleum refining and its methods, refining capacity and consumption in India, constituents of Crude Oil, Recovery techniques – Oil – Gas separation, Processing wet gases.

P & I Diagrams for Petrochemical Plants: P & I diagram of petroleum refinery, Atmospheric distillation process, Vacuum distillation process, Thermal cracking, Catalytic cracking, Catalytic reforming, and Utility plants – Air, N₂, and cooling water.

UNIT-II:

Instruments for Petrochemical Industries: Basics of field instruments, Parameters to be measured in the Petrochemical industry, Distillation Column control, Selection of instruments, intrinsic safety of instruments, and Area classification.

UNIT-III:

Petroleum and its Chemicals: Chemicals from petroleum, Chemicals from methane, acetylene, ethylene and propylene - production routes of important petrochemicals such as polyethylene, polypropylene, ethylene dioxide, methanol, xylene, benzene, toluene, styrene, VCM and PVC.

UNIT-IV:

Control of Petroleum Industries and its Units: Control of furnace, Reboiler Control, Reflux Control, Control of catalytic crackers, Control of heat exchanger, Control of cooling tower.

UNIT-V:

Safety Systems for Petroleum Industries: Basics of PLC and Safety interlocks in furnace, separator, pump, and compressor. Basics of SIL, Introduction to Standards.

TEXT BOOKS:

1. Chemical from Petroleum, Wadams A. L., Butter and Janner Ltd., Chemical Pub Publication, 1969
2. Process Control Structures and Applications, Balchan J. G. and Mumme K. I., Van Nostrand, Reinhold Company, 1988
3. Chemical Process Industries, Austin G. T. Shreves, McGraw-Hill International, 1985

REFERENCES:

1. Instrumentation in Process Industries, Bela G. Liptak, Chilton Book Company, 2005
2. Oil and Gas Production Handbook-An Introduction to Oil and Gas Production, Havard Devold, ABB ATPA Oil and Gas, 2015

B.Tech. VII Semester

(22PE1EC407) SPEECH SIGNAL PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Signals and Systems, Digital Signal Processing

COURSE OBJECTIVES

- To understand speech production and related parameters of speech
- To learn the time domain analysis of speech signal
- To understand the spectral analysis of speech and modeling
- To learn the enhancement and coding of speech signal
- To study speech recognition and synthesis system

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic organs and parameters of speech

CO-2: Analyze the speech signal in time domain

CO-3: Probe the speech spectrum and modelling

CO-4: Enhance and code the speech signal

CO-5: Design a speech recognition and synthesis system

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	-	-	-	-	-	-	-	-	-	1	1	1
CO-2	3	3	2	-	-	-	-	-	-	-	-	-	2	2	2
CO-3	3	3	2	-	-	-	-	-	-	-	-	-	2	-	2
CO-4	3	3	3	-	-	-	-	-	-	-	-	-	2	-	2
CO-5	3	2	3	-	-	-	-	-	-	-	-	-	2	-	2

UNIT-I:

Fundamentals of Speech Production and Processing: Anatomy and physiology of speech organs, the process and acoustic theory of speech production, uniform lossless tube model, effect of losses in vocal tract, effect of radiation at lips, digital models for speech signals.

UNIT-II:

Time Domain Methods for Speech Processing: Time domain parameters of speech signal, methods for extracting the speech parameters, short time auto correlation function, pitch period estimation using auto correlation function.

UNIT-III:

Frequency Domain Methods for Speech Processing: Short-time Fourier transform (STFT), sampling the STFT in time and frequency, the speech spectrogram, homomorphic

systems for convolution, definition of the Cepstrum and complex Cepstrum, pitch extraction using homomorphic speech processing.

Speech Modeling: Introduction to Markov Processes, Hidden Markov Model development and Evaluation, Optimal State Sequence Viterbi Search algorithm, Baum-Welch Parameter Re-estimation.

UNIT-IV:

Speech Enhancement: Introduction, Single Microphone Approach, Spectral Subtraction, Enhancement by re-synthesis, Comb filter, Wiener filter, Multi Microphone Approach.

Speech Coding: Predictive Coding, Delta Modulation, Adaptive Differential PCM Systems, Multi-Pulse Excitation Linear Prediction (MPLP), Code-Excited Linear Prediction (CELP).

UNIT-V:

Speech Recognition: Architecture of large vocabulary continuous speech recognition system, acoustics and language models

Speech Synthesis: Concatenative and waveform synthesis methods, Intelligibility and naturalness.

TEXT BOOKS:

1. Introduction to Digital Speech Processing, Lawrence R. Rabiner and Ronald W. Schafer, now Publishers Inc., 2007
2. Speech Enhancement, Philipos C. Loizou, 2nd Edition, CRC Press, Taylor & Francis, 2013
3. Fundamentals of Speech Recognition, Lawrence Rabiner and Biing-Hwang Juang, Pearson Education, 2003

REFERENCES:

1. Digital Processing of Speech Signals, L. R. Rabiner and R.W. Schafer, Prentice-Hall, 1978
2. Speech Communications Human and Machine, Douglas O. Shaughnessy, 2nd Edition, IEEE Press, 2000
3. Statistical Methods of Speech Recognition, Frederick Jelinek, MIT Press, 1997
4. Speech Recognition, Claudio Becchetti and Lucio Prina Ricotti, John Wiley and Sons, 1999
5. Discrete Time Speech Signal Processing: Principles and Practice, Thomas F. Quateri, Pearson Education, 2004

B.Tech. VII Semester

(22PE1EC421) WIRELESS SENSOR NETWORKS AND PROTOCOLS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Computer and Communication Networks

COURSE OBJECTIVES:

- To understand basics of Wireless Sensor Networks
- To study of medium access control protocols
- To distinguish key routing protocols used in sensor networks
- To learn transport layer protocols used in for sensor networks

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify engineering components, systems and protocols necessary for establishing the WSN

CO-2: Demonstrate proficiency in using different WSN Protocols

CO-3: Identify the design goals and issues in Routing and Transport Layer Protocols for Ad Hoc Wireless Networks

CO-4: Address the security threats in Wireless Sensor Networks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	2	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	3	2	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	2	2	2	-	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Overview of Wireless Sensor Networks: Key definitions of sensor networks, Advantages of sensor Networks, Unique constraints and challenges, Driving Applications.

Networking Technologies: Physical Layer and Transceiver Design Considerations, Personal Area Networks (PANs), Hidden node and exposed node problem, Topologies of PANs, MANETs, WANETs.

UNIT-II:

MAC Protocols for Wireless Sensor Networks: Issues in Designing a MAC protocol for Ad Hoc Wireless Networks, Design goals, Classifications of MAC Protocols, MAC Protocols that use Directional Antennas.

UNIT-III:

Routing Protocols: Introduction, Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks, Classification of Routing Protocols, Table-Driven Routing Protocols,

On-Demand Routing Protocols, Hybrid Routing Protocols, Routing Protocols with Efficient Flooding Mechanisms.

UNIT-IV:

Transport Layer Protocols: Introduction, Issues in Designing a Transport Layer Protocol for Ad Hoc Wireless Networks, Design Goals of a Transport Layer Protocol for Ad Hoc Wireless Networks, Classification of Transport Layer Solutions.

UNIT-V:

Security in WSN: Security in Ad Hoc Wireless Networks, Network Security Requirements, Issues and Challenges in Security Provisioning, Network Security Attacks, Key Management, Secure Routing in Ad Hoc Wireless Networks.

TEXT BOOKS:

1. Adhoc Wireless Networks: Architectures and Protocols, C. Siva Ram Murthy and B. S. Manoj, PHI, 2004
2. Wireless Adhoc and Sensor Networks: Protocols, Performance and Control, Jagannathan Sarangapani, CRC Press, 2007
3. Protocols and Architectures for Wireless Sensor Networks, Holger Karl & Andreas Willig, John Wiley, 2005

REFERENCES:

1. Wireless Sensor Networks - Technology, Protocols, and Applications, Kazem Sohraby, Daniel Minoli & Taieb Znati, John Wiley, 2007
2. Wireless Sensor Networks - An Information Processing Approach, Feng Zhao & Leonidas J. Guibas, Elsevier, 2007
3. Adhoc Mobile Wireless Networks: Protocols & Systems, C. K. Toh, 1st Edition, Pearson Education
4. Wireless Sensor Networks, C. S. Raghavendra, Krishna M. Sivalingam, Springer, 2004
5. Wireless Sensor Networks, S. Anandamurugan, Lakshmi Publications

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EI403) BIOMEDICAL SIGNAL PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To interpret the essential bio signals such as ECG and EEG
- To apply signal and data processing techniques to bio signals and applications in biomedicine
- To illustrate the use of wavelets in medical applications
- To grasp the advancements of biomedical engineering with the help of emerging technologies like BCI

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Reflect on biological systems from a signals and systems viewpoint and apply suitably signal processing techniques

CO-2: Apply advanced data compressing, modelling, and signal processing techniques to ECG and EEG signals

CO-3: Design and implement digital filters for noise reduction in electrophysiological data

CO-4: Demonstrate real-world applications of BCI

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	1	-	1	1	-	-	1	2	-	1	1	2	1
CO-2	1	2	1	-	2	2	-	3	1	3	-	1	1	2	1
CO-3	1	1	1	-	1	3	-	2	2	3	-	2	3	2	2
CO-4	2	2	1	-	2	1	-	2	1	2	-	3	3	2	2

UNIT-I:

Cardiological Signal Processing: preprocessing of ECG signal, QRS detection methods-Differentiation based, template based, Rhythm analysis and Arrhythmia detection algorithms. Automated ECG analysis.

UNIT-II:

Data Compression Techniques: Turning Point algorithm, AZTEC, CORTES, KL transform, Adaptive filters, Weiner filter principles, LMS & RLS, medical Applications of Adaptive Noise Cancellation.

UNIT-III:

Neurological Signal Processing: Stochastic process, Linear prediction, Yule-Walker equations, Auto Regressive Modeling of EEG signal, Detection of EEG Rhythms, Template matching for EEG spike and wave detection, Detection of EEG spike and

wave complexes, Coherence analysis of EEG channels, Adaptive segmentation of EEG signals.

UNIT-IV:

SLEEP EEG: Data Acquisition and Classification of Sleep stages, The Markov Model and Markov Chains, Dynamics of Sleep-Wake Transitions, Hypnogram Model Parameters. PRONY'S Method: Exponential Modelling, Exponential Parameter Estimation, The original Prony Problem, Least Squares Prony Method, The Covariance Method of Linear Prediction.

UNIT-V:

Wavelets in Medicine: Need for wavelets, Types of wavelets, Selection of a wavelet for an application, Decomposition and reconstruction of signals using wavelets, Denoising using wavelets, typical medical applications.

Brain-Computer Interface: Brain signals for BCIs, Generic setup for a BCI, Feature extraction and Feature translation involved in BCIs. Typical medical applications.

TEXT BOOKS:

1. Biomedical Signal Analysis: A Case-Study Approach, Rangaraj M. Rangayyan, John Wiley & Sons, 2009
2. Biomedical Signal Processing, D. C. Reddy, The McGraw-Hill Companies, 2005

REFERENCES:

1. Biomedical Digital Signal Processing, Willis J. Tompkins, Prentice-Hall of India, 2012
2. Statistical Digital Signal Processing and Modeling, Monson H. Hayes, Wiley-India, 2009
3. Brain-Computer Interfaces: Principles and Practice, Jonathan Wolpaw and Elizabeth Winter Wolpaw, Oxford University Press, 2012
4. Wavelet Tour of Signal Processing: The Sparse Way, Stephan Stephane Mallat, 3rd Edition, Academic Press, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EI401) IOT APPLICATIONS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To understand embedded system programming and design and develop embedded solutions
- To impart system level thinking
- To acquire innovative design applications
- To acquire command over design automation

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Acquire knowledge with programming environment to develop Embedded and IoT Applications

CO-2: Implement Interfacing of input-output modules to various processors

CO-3: Design & Implement Small Embedded applications using controllers, Sensors, and Actuators

CO-4: Design & Implement Small IoT applications using controllers, Sensors, and Actuators

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	1	1	3	3	-	-	-	-	-	3	3	3	2
CO-2	3	1	1	1	3	3	-	-	3	3	3	3	3	2	2
CO-3	3	2	2	2	3	3	-	-	3	3	3	3	3	2	2
CO-4	3	3	3	2	2	2	-	-	3	3	3	3	3	2	2

LIST OF EXPERIMENTS:

Interfacing:

1. Interfacing of Keypad and LCD with ARM Processors
2. Interfacing of GPS & GSM with ARM Processors

Embedded Applications (with ARM Processors /Arduino):

3. Implementation of Multichannel Data Acquisition Systems.
4. Automatic Wiper control in Automobile System.
5. Implementation of Smart Traffic Management System.
6. Implementation of Multilevel Vehicle Parking System.
7. Implementation of Line Follower Robot.
8. Implementation of Obstacle Avoidance Robot.

IoT Applications (with Arduino / Raspberry Pi):

9. Design and development of Home Automation System.

10. Design and Control of Smart Irrigation System.
11. Design and development of a Weather reporting system.
12. Implementation of health monitoring System.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EI402) ANALYTICAL INSTRUMENTATION LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To introduce the whole array of modern analytical instruments to implement statical analysis tools
- To emphasize a hands-on approach with sample preparation, application, method development, data analysis and interpretation being key elements
- To introduce methods to Interpret data derived from any analytical instrument
- To familiarize with the basic concepts, principles and terms of chromatography, Spectroscopy, and gas analyzers

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify modern analytical instruments to analyze liquids, solids, and gases samples

CO-2: Perform a hands-on approach with sample preparation, method development, measurement, data analysis, interpretation and application

CO-3: Apply the basic concepts and principles of chromatography, spectroscopy and gas analyzers to measure the concentration of different constituents

CO-4: Measure and analyze the nuclear radiation intensity of radioactive materials

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	2	-	2	1	-	1	2	1	-	1	1	1	-
CO-2	1	3	1	-	1	2	-	2	2	2	-	2	1	2	-
CO-3	2	1	2	1	2	2	-	2	3	3	-	2	1	3	-
CO-4	2	1	2	-	1	1	-	2	1	2	-	2	1	2	-

LIST OF EXPERIMENTS:

- Analysis of CO₂, O₂, HC and CO concentration in the ambient air & automobile emissions using a gas analyzer.
- Separation of different constituents in a mixture of chemicals using chromatography.
- Identification of atoms and its concentration through absorption spectra with UV-VIS spectrophotometer.
- Identification of chemical compounds and its concentration using FTIR spectrometer.
- Identification of atoms and its concentration through emission spectra using a flame photometer.
- Measurement of calorific value using a digital bomb calorimeter.
- Determination of the acid/alkaline nature of water using a pH meter.

8. Qualitative and quantitative analysis of milk and protein using Milk Analyzer.
9. Blood glucose analysis using a glucometer.
10. Radiation intensity measurement with varying distance and measurement of absorber thickness using nuclear radiation detector-GM counter.
11. Analysis of water quality using a water purity meter.
12. Measurement of total dissolved salts and water conductivity using a digital conductivity meter.
13. Measurement of turbidity of water using a digital turbidity meter.
14. Measurement of the color concentration of a sample using a microprocessor-based photo calorimeter.

B.Tech. VIII Semester

(22PE1EI404) ROBOT PROGRAMMING AND INTELLIGENT CONTROL SYSTEM

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the fundamentals of robot programming
- To learn robot textual languages that are in common use
- To expose to artificial intelligence in robotics
- To acquire basic Knowledge of neural networks in robotics
- To acquire basic Knowledge of fuzzy logic in robotics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain knowledge about different methods of robot programming.

CO-2: Examine various robot language elements and their functions.

CO-3: Analyze different AI techniques employed in robotics.

CO-4: Design basic neuro-controller for robot motion control.

CO-5: Apply fuzzy logic to robot control systems.

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	-	-	1	-	-	-	-	1	-	2	2	1	3
CO-2	3	1	2	-	1	-	-	-	-	1	-	2	2	2	3
CO-3	3	2	2	-	2	-	-	-	-	1	-	2	3	2	3
CO-4	3	2	3	2	2	-	-	-	-	-	-	-	2	1	2
CO-5	3	3	2	2	2	-	-	-	-	-	-	-	2	1	2

UNIT-I:

Robot Programming: Methods of robot programming, lead through programming methods, robot program as a path in space - defining position in space, speed control, motion interpolation, WAIT, SIGNAL, DELAY commands, Branching.

Robot Languages: Textual robot language, generations of robot languages, robot language structure, operating systems, Robot Language Elements and functions, constraints and variables, aggregates and location variables.

UNIT-II:

Basic Commands and Operations: Motion commands- move and related statements, speed control, points in workspace, paths and frames. End effector and sensor commands- end effector operation, sensor operation, REACT statement.

Computations and operation. Program control and subroutines. Communications and data processing. Monitor mode commands.

UNIT-III:

AI for Robotics: Introduction to Artificial Intelligence, goals of AI research, AI techniques- knowledge representation, problem representation, search techniques. LISP programming. AI and Robotics. LISP in the factory. Robotic Paradigms.

UNIT-IV:

Neural Network Approach in Robotics: Introduction, Connectionist Models, Learning Principles and Learning Rules: Supervised, unsupervised, reinforcement learning. Sensor based robot learning, Neural Network in Robotics: Control of robot hands by neural network, neural set approach to robot motion coordination, robotic motor control using reinforcement learning optimization.

UNIT-V:

Fuzzy Logic Approach in Robotics: Introduction, Fuzzy sets, Operation of Fuzzy sets, Fuzzy relations, Fuzzy rule formation, Control rules, Fuzzy algorithm in robotics, Robot obstacle avoidance using fuzzy logic, Fuzzy logic for robot path tracking and behaviour coordination, fuzzy control system in mobile robots, fuzzy controller design for robot systems, Case study of fuzzy logic in robotics.

TEXT BOOKS:

1. Industrial Robotics Technology, Programming and Applications, Mikell P. Groover, McGraw-Hill, 2012
2. Robotics Technology and Flexible Automation, Deb S. R., 2nd Edition, Tata McGraw-Hill, 2017

REFERENCES:

1. Design and Control of Intelligent Robotic Systems, (Studies in Computational Intelligence 177) M. Begum, F. Karray, Dikai Liu, Lingfeng Wang, Kay Chen Tan (Eds.), Springer, 2010
2. Neural Networks in Robotics, Edited by George Bekey, Kenneth Y. Goldberg, Springer, 2012
3. Neural Networks, Fuzzy Logic, Genetic Algorithm - Synthesis and Applications, Rajasekharan and Rai, 1st Edition, PHI Publications, 2013
4. Introduction to Neural Networks using MATLAB 6.0, S. N. Sivanandam, S. Sumathi, S. N. Deepa, 1st Edition, McGraw-Hill Education, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EI405) INSTRUMENTATION FOR PHARMACEUTICAL INDUSTRIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn the sequence of operations in the pharmaceutical industry
- To know the necessity of precision instrumentation in the pharmaceutical industry
- To study the different processes that are performed in the pharmaceutical industry
- To know the different components and equipment required to process and control various parameters in the pharmaceutical industry

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and interpret various filters and centrifuge devices in the pharmaceutical Process applications

CO-2: Analyze the instrumentation for crystallization, vaporization and distillation process in pharmaceutical process applications

CO-3: Analyze the procedure to control humidity and refrigeration, crystallization, and vaporization

CO-4: Interpret different methods and devices used for size reduction, separation, and homogenization in pharmaceutical process applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	-	-	-	-	2	2	-	1	1	1	1	-	1	1
CO-2	1	1	2	1	-	2	2	-	1	1	1	1	-	1	1
CO-3	1	1	-	1	-	1	2	-	1	1	1	1	-	1	-
CO-4	1	-	1	1	-	1	1	-	1	-	-	1	-	-	-

UNIT-I:

Introduction to Pharma Industries: Basic Processors and Instrumentation Techniques, Process Analysis Technology (PAT). Filtration: Classification of Filtration, Mechanism of Filtration, Filter media, Filter Aids, Pre-treatment of materials, small scale filtration methods, filtration equipment, filter presses, Leaf filters, stacked disc filters, meta filters, Rotary continuous filters, other methods, ceramic filters, Seitz filters, sintered (fritted) Glass filters, Membrane filters, factors affecting the rate of filtration, industrial filters including rotary filter, edge filter.

Centrifugation: General principles, theoretical aspects, classification, Laboratory equipment, Large scale equipment, Semi-continuous centrifuge, equipment with non-

perforated basket, de Laval clarifier, vertical solid bowl centrifuges, and continuous centrifuges.

UNIT-II:

Crystallization: Introduction, Crystal forms and crystal Habit, classification of crystallizers, tank crystallizers, agitated batch crystallizers, Swenson Walker Crystallizer, Krystal Crystallizer, Vacuum Crystallizer without External Classifying seed Bed, theoretical aspects of Crystallization, Calculation of yields and theory of Crystallization.

UNIT-III:

Evaporation and Distillation Heat Processes: Factors affecting evaporation, a study of evaporating stills and evaporating pans, heat transferring evaporators, vapour compression evaporators and evaporation under reduced pressure.

Distillation: The importance of distillation in pharmacy and the methods of distillation. A brief introduction to freeze drying, sublimation, desiccation and exsiccation, efflorescence and its importance.

Humidity Control and Refrigeration: Basic concepts and definition, wet bulb temperature, adiabatic cooling lines, use of Humidity chart, determination of Humidity, air conditioning, humidification and humidifying equipment, humidifiers. Introduction: refrigeration equipment, coefficient of performance and refrigerants, brine systems, refrigeration load, and absorption systems.

UNIT-IV:

Size Reduction and Separation: Introduction, mechanism and principles of size reduction, classification of size reduction equipment, law of size reduction, large equipment, mills using impact force for size reduction, cage mills, pin mills, fluid energy or jet mills, attrition and grinding mills, tumbling mills.

Size separation sieving, Screening equipment, sedimentation, screen analysis Definition, objectives of size reduction, factors affecting size reduction, laws governing energy and power requirements of a mill, types of mills including ball mill, hammer mill, fluid energy mill, etc.

UNIT-V:

Mixing and Homogenization: Introduction, equipment for mixing miscible liquids, mixing of a soluble solid with a low viscous liquid., mixing solids with solids, equipment, consideration while choosing solids mixing equipment, mixing solids with liquids, mixing miscible liquids, mixing viscous masses, mixing of immiscible liquids, equipment for emulsification.

TEXT BOOKS:

1. Pharmaceutical Engineering, K. Samba Murthy, 1st Edition, New Age Publishers, 1998
2. Pharmaceutical Engineering: Unit Operations, C. V. S. Subhramanyam, 4th Edition, Vallabh Prakashan, 2022
3. Tutorial Pharmacy, S.J. Carter, Cooper and Gunn's, 6th Edition, CBS Publisher, 2005

REFERENCES:

1. Perry's Handbook of Chemical Engineering, 9th Edition, McGraw-Hill, 2019

B.Tech. VIII Semester

(22PE1EI406) NEURAL NETWORKS AND FUZZY LOGIC SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the fundamental concepts of neural networks, including their architectures, models, and learning methods
- To explore the principles of backpropagation neural networks and their learning algorithms for problem-solving
- To examine associative memory networks and their applications in storage and recall mechanisms
- To understanding and designing Convolution Neural Networks (CNNs)
- To gain comprehensive knowledge of fuzzy logic systems, their operations, and real-time applications using fuzzy and neuro-fuzzy control techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate an understanding of the fundamental concepts of neural networks, including architectures and learning methods

CO-2: Apply backpropagation algorithms for neural network training and evaluate the influence of parameters on network performance

CO-3: Design and analyze associative memory networks, CNNs and their stability for various applications

CO-4: Design fuzzy logic-based systems and apply them to real-time scenarios like sensor fault detection, automation, and dynamic control

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	1	2	1	1	2	1	1	1	2	3	2	1
CO-2	3	3	3	2	3	2	1	2	1	1	2	2	3	3	2
CO-3	3	3	3	3	3	2	3	2	2	1	2	3	3	3	3
CO-4	3	3	3	2	3	3	3	2	2	1	2	3	3	3	3

UNIT-I:

Fundamentals of Neural Networks: Basic concepts of neural networks, Human brain, model of an artificial neuron, Neural network architectures: Single layer feedforward network, Multilayer feedforward network, Characteristics of neural networks, McCulloch-Pitts Model, learning methods, Taxonomy of neural network architectures, early neural network architectures: Rosenblatt's Perceptron, Perceptron and linearly separable tasks, XOR problem.

UNIT-II:

Back Propagation Networks: Back propagation architecture, Back propagation algorithm, Back propagation learning: Input layer computation, hidden layer computation, output layer computation, calculation of error, generalized delta rule, Training of the neural network, Method of steepest descent, The effect of learning rate ' η ', Selection of various parameters in BPN

UNIT-III:

Associative Memories: Auto Associative memory, Hetero Associative Memory Neural Network, Bidirectional Associative Memory (BAM) Architecture, BAM training algorithms: storage and recall algorithm, Autocorrelators, BAM energy function, LSTMs.

Convolutional Neural Networks (CNNs): Basics of CNN pooling and architecture, Designing CNNs from fully connected layers, Training CNN, and Model optimization.

UNIT-IV:

Fuzzy Logic: Introduction to Fuzzy Logic. Fuzzy sets and membership functions. Operations on Fuzzy Sets. Fuzzy relations, rules, propositions, implications, and inferences. Defuzzification techniques. Some applications of Fuzzy logic.

UNIT-V:

Applications: Fuzzy Logic models for dynamic systems, Sensor Fault Detection, CNNs for Soil Classification and Facial Recognition, Fuzzy and Neuro-Fuzzy control schemes: Temperature and Level Control Applications.

TEXT BOOKS:

1. Neural Networks, Fuzzy Logic, and Genetic Algorithms: Synthesis and Applications, R. Rajasekaran and G.A. Vijayalakshmi, Prentice Hall of India, 2003
2. Soft Computing, D. K. Pratihar, Narosa Publishing, 2014

REFERENCES:

1. Fuzzy Logic with Engineering Applications, Timothy J. Ross, 3rd Edition, Wiley, 2011
2. Introduction to Artificial Neural Networks, M. Paul Raj & S. N. Sivanandam, Vikas Publishing, 2003
3. Neural Networks, Simon Haykins, Pearson Education, 3rd Edition, 2008
4. Practical Convolutional Neural Networks, Mohit Sewak, Md. Rezaul Karim, Pradeep Pujari, Packt Publishing, 2018
5. Guide to Convolutional Neural Networks, Hamed Habibi Aghdam, Elnaz Jahani Heravi, Springer International, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC402) DSP PROCESSORS AND ARCHITECTURES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Digital Signal Processing, Microprocessors and Microcontrollers

COURSE OBJECTIVES:

- To study the architectural features of DSP Processor
- To learn about numeric formats and sources of errors in DSP implementation
- To understand the basic building blocks of DSP Processor
- To understand the code optimization techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the architecture and evaluate the finite word length effects

CO-2: Apply the pipelining concepts and evaluate the computational accuracy

CO-3: Utilize the basic building blocks of DSP processor

CO-4: Implement the real time application using DSP processor

CO-5: Apply the Code Optimization Techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	-	-	-	-	-	-	-	-	3	1	2
CO-2	3	2	2	-	-	-	-	-	-	-	-	-	2	1	2
CO-3	3	2	2	-	-	-	-	-	-	-	-	-	3	1	2
CO-4	3	2	2	-	-	-	-	-	-	-	-	-	3	1	2
CO-5	3	2	2	-	-	-	-	-	-	-	-	-	2	1	2

UNIT-I:

Introduction to DSP Processors: Digital Signal Processors, VLIW and SHARC Architecture, Comparison of DSP processors with microprocessor

Implementation Considerations: Data representations and arithmetic, Introduction to finite word length effects, Input quantization, Coefficient quantization, overflow and solutions, rounding and truncation, real time implementation considerations.

UNIT-II:

Execution Control and Pipelining: Hardware looping, Interrupts, Stacks, Relative Branch support, Pipelining and Performance, Pipeline Depth, Interlocking, Branch effects, Interrupt effects, Pipeline Programming models.

Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP system, Dynamic range and precision, sources of errors in DSP implementation, A/D conversion errors, DSP computational errors, D/A conversion errors.

UNIT-III:

Architectures for programmable DSP Devices: Basic Architectural features, DSP computational building blocks, bus architecture and memory, Data and addressing capabilities, Address generation unit, programmability and program execution, speed issues, Features for external interfacing.

UNIT-IV:

Architecture and Instruction Set of the C6x Processor: TMS320C6x Architecture, Functional Units, Fetch and Execute Packets, Pipelining, Registers, Linear and Circular Addressing Modes, TMS320C6x Instruction Set

On-chip Peripherals: Timers, Interrupts, Interrupt Control Registers, Interrupt Acknowledgment, Multichannel Buffered Serial Ports, Direct Memory Access, Memory Considerations

UNIT-V:

Code Optimization: Optimization Steps, Compiler Options, Intrinsic C Functions, Procedure for Code Optimization, Programming Examples Using Code Optimization Techniques

TEXT BOOKS:

1. Digital Signal Processing, Avtar Singh and S. Srinivasan, Thomson Publications, 2016
2. Digital Signal Processing - A Practical Approach, Emmanuel C. Ifeachor, Barrie W. Jervis, 2nd Edition, Pearson Publications, 2002
3. Digital Signal Processing and Applications with the C6713 and C6416 DSK, Rulph Chassaing, Wiley, 2005

REFERENCES:

1. Digital Signal Processors Architectures, Implementations and Applications, Sen M.Kuo, Woon-Seng S. Gan, Pearson Publications, 2009
2. Digital Signal Processors, Architecture, Programming and Applications, B. Venkata Ramani and M. Bhaskar, Tata McGraw-Hill, 2007
3. DSP Processor Fundamentals, Architectures and Features, Lapsley, S. Chand, 2003
4. DSP Applications with TMS 320 Family, K. Shin, Prentice Hall, 1987

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EI407) TELE MEDICINE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the evolution and significance of telemedicine
- To examine the role of RFID and other wireless technologies in enhancing telemedicine practices
- To know different technologies used in healthcare and the importance of safeguarding medical data
- To identify and analyze emerging trends in telemedicine, including home care technologies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze the historical and modern developments in telemedicine and e-health systems

CO-2: Evaluate the effectiveness of various wireless networks and RFID technologies in healthcare environments

CO-3: Utilize medical information processing techniques for data collection, bio-signal processing and patient record management and safeguarding

CO-4: Predict and evaluate future advancements in telemedicine and their implications for healthcare delivery systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	2	2	1	1	2	2	1	1	1	1	3	1	3	1
CO-2	1	2	2	2	2	2	2	1	1	1	2	3	1	3	1
CO-3	1	3	3	2	2	3	2	1	1	1	2	3	1	3	1
CO-4	1	3	2	2	1	3	2	1	1	1	2	2	1	3	1

UNIT-I:

Introduction: Information Technology and Healthcare Professionals; Providing Healthcare to Patients -Technical Perspective, Healthcare Providers, End Users, Authorities; Healthcare Informatics Developments; Different Definitions of Telemedicine; Overview on Telemedicine, The Growth of the Internet: Information Flooding in E-Health.

UNIT-II:

Communication Networks and Services: Wireless Communications Basics; Types of Wireless Networks; The Outdoor Operating Environment; RFID in Telemedicine

UNIT-III:

Technologies in patient monitoring and information processing: Wireless Technology in Patient Monitoring-Body Area Networks, Emergency Rescue, and Remote Recovery, At the Hospital.

Technologies in Medical Information Processing- Collecting Data from Patients; Bio-signal Transmission and Processing; Patient Records and Data Mining Applications; Knowledge Management for Clinical Applications; Electronic Drug Store

UNIT-IV:

Technologies for Safeguarding Medical Data and Privacy: Technologies for Safeguarding Medical Data and Privacy; Cryptography; Safeguarding Patient Medical History; Anonymous Data Collection and Processing; Biometric Security and Identification

UNIT-V:

Future Trends in Healthcare Technology: Prognostics in Telemedicine; The Aging Population: Home Care for the Elderly; Clothing Technology and Healthcare; Haptic Sensing for Practitioners; The Future of Telemedicine and Information Technology for Everyone.

TEXT BOOK:

1. Telemedicine Technologies Information Technologies in Medicine and Telehealth, A. C. M. Fong, C. K. Li,

REFERENCES:

1. The Natural Radiation Environment, US Department of Energy Nuclear Testing Archive, Adams, J. A. S., Louder, W. M., Phair, G., and Gottfried, D., 1964
2. Essentials of Telemedicine and Telecare, A. C. Norris, John Wiley & Sons, 2002

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EI408) MICRO ELECTRO-MECHANICAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To acquire knowledge about MEMS devices and their applications in various domains
- To understand the techniques to fabricate MEMS devices
- To learn the design considerations for MEMS devices and microsystems
- To learn to characterize Microsystems using optical and electron microscopy and other techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Appraise the application of various materials commonly used for MEMS fabrication

CO-2: Select appropriate tools and techniques for microfabrication and MEMS characterization

CO-3: Realize the need for advancement of technology towards Microsystems for better living in society

CO-4: Understand the need to keep oneself updated constantly to understand the ease of use of emerging technologies

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	-	3	1	-	1	2	2	-	1	1	2	-
CO-2	2	2	1	-	2	2	2	2	2	2	-	3	1	2	-
CO-3	2	3	2	1	2	2	-	3	3	3	-	2	1	3	-
CO-4	2	2	2	-	2	1	-	2	1	2	-	3	1	1	-

UNIT-I:

Overview of MEMS and Microsystems: Evolution of microfabrication, Applications of MEMS in optical devices (Micro-Opto-Electro- Mechanical Systems or MOEMS), healthcare and biomedicine (including Bio-MEMS and Bio-MOEMS), aerospace, telecommunications, consumer products, automotive, and industrial products.

UNIT-II:

Working Principles of Microsystems: Microsensors–acoustic wave, biochemical, optical, pressure, thermal, Microactuation – thermal, shape-memory alloys,

piezoelectric, electrostatic, MEMS devices – Microgrippers, Micromotors, Microfluidics– Micropumps, Microvalves, Micro accelerometers.

UNIT-III:

Materials for MEMS and Microsystems: Substrates and Wafers, Silicon as a Substrate, Silicon Compounds, Silicon piezo resistors, non-silicon based materials-Gallium Arsenide, Gallium Nitride, Quartz, Piezoelectric Crystals, Polymers.

UNIT-IV:

Basics of Micromanufacturing: Photolithography, Cleanroom Environment, Deposition techniques-Ion implantation, Diffusion, Vapour Deposition (PVD, CVD, PECVD), Oxidation, Epitaxial growth, Etching techniques-Chemical (Wet) Etching, Plasma (Dry) Etching

UNIT-V:

Fabrication of MEMS: Bulk micromachining, Surface micromachining, LIGA Process, Deep X-ray lithography (DXRL), Design considerations, Process Design, Photomask layout using CAD, Mechanical design overview.

Characterization of MEMS: Characterization Techniques, Principle of working and operation of Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM), Atomic Force Microscope (AFM), X-ray diffraction (XRD), Optical microscope

TEXT BOOKS:

1. MEMS and Microsystems Design and Manufacture, Tai-Ran Hsu, 2nd Edition, Tata McGraw-Hill, 2002
2. MEMS, N. Mahalik, 1st Edition, McGraw-Hill Education, 2007

REFERENCES:

1. Fundamentals of Microfabrication: The Science of Miniaturization, Marc J. Madou, 3rd Edition, CRC Press, 2002
2. Microsystem Design, Stephen D. Senturia, 1st Edition, Springer, 2004
3. Practical MEMS, Ville Kaajakari, 2nd Edition, Small Gear Publishing, 2009

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EI409) POLLUTION CONTROL IN PROCESS INDUSTRIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify the different pollutants that cause serious problems in industries
- To understand the causes of pollution from various emission reactions
- To recognize the origin of hazardous pollution gases in the environment
- To Treatment methods for specific pollutants arising from industrial processes

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Plan strategies to control and reduce pollution

CO-2: Select the most appropriate techniques to control and treat industrial pollution

CO-3: Apply environmental management systems (EMS) to an industrial activity

CO-4: Design and develop anti-pollution monitoring systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	1	1	2	1	1	-	1	-	1	1	3	3
CO-2	1	1	2	1	1	2	2	1	-	-	1	2	1	3	3
CO-3	1	1	1	2	2	2	2	1	1	1	-	1	1	3	2
CO-4	2	1	3	2	1	1	2	1	1	-	2	1	1	3	2

UNIT-I:

Industrial Pollution Emissions and Indian Standards: Introduction- Man and environment, types of pollution, pollution control aspects, Industrial Emissions-Gases, Industrial emissions Liquids, water quality management in India.

UNIT-II:

Analysis of Pollutants: Industrial waste water analysis, industrial gaseous effluent analysis, particle size distribution, water quality regulations and policy development, water quality standards.

UNIT-III:

Pollution Control for Specific Pollutants: Removals of BOD-biological oxidation anaerobic treatment-Removal of Chromium-control methods, reduction precipitation, Ion exchange, reverse osmosis-lime coagulation and adsorption.

Removal of mercury, measurement of mercury, mercury losses in chloro-alkali industries, removal of mercury from gaseous streams, removal of mercury from liquid streams, Removal of nitrogen oxides, analysis of NO_x control measures.

UNIT-IV:

Pollution Control Aspects in Selected Process Industries: Pollution control in chemical industries, pollution control aspects of fertilizer industries, ammonia plant effluents, ammonium sulphate plants, phosphoric acid plants, and complex fertilizer plants.

UNIT-V:

Pollution Control in Petroleum Refineries and Petrochemical Units: Characteristics of liquid effluent-refinery, liquid waste treatment methods, treatment of liquid effluents from petrochemical industries, air pollution control, pollution control in pulp and paper industries.

TEXT BOOKS:

1. Pollution Control in Process Industries, S. P. Mahajan, 32nd Edition, Tata McGraw-Hill, 2017
2. Water Quality Concepts, Sampling, and Analyses, Yuncong Li, Kati Migliaccio, 1st Edition, CRC Press, 2010

REFERENCES:

1. Industrial Pollution Prevention Handbook, 1st Edition, Harry M. Freeman, McGraw-Hill Education, 2017
2. Industrial Pollution (A Reference to Small Scale Industries), 1st Edition, N. Saradha, N. Dhulasi Birundha, Serials Publications, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC305) DIGITAL IMAGE PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Digital Signal Processing

COURSE OBJECTIVES:

- To introduce fundamentals of digital image processing and study image transforms
- To learn enhancement techniques in spatial and frequency domains
- To study and compare various image compression, image segmentation and morphological algorithms
- To understand image representation methods

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand the basic principles of digital image processing and perform image transforms

CO-2: Perform basic image processing methods such as Image filtering operations, Image enhancement

CO-3: Segment the image and apply wavelet transforms

CO-4: Compress the image and apply Morphological operations on image

CO-5: Describe and represent the image for computer processing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	1	1	-	1	-	1	-	-	1	3	2
CO-2	2	2	2	1	1	1	-	1	-	1	-	-	1	3	2
CO-3	3	2	3	1	1	1	-	1	-	1	-	-	1	3	2
CO-4	2	2	2	1	1	1	-	1	-	1	-	-	1	3	2
CO-5	3	3	1	1	1	1	-	1	-	1	-	-	1	3	2

UNIT-I:

Fundamentals of Image Processing: Digital Image Fundamentals, Basic steps of Image Processing System, Sampling and Quantization of an image, relationship between pixels, Imaging Geometry.

Image Transforms: 2 D- Discrete Fourier Transform, Discrete Cosine Transform (DCT), Haar Transform, Hadamard Transform, Hotelling Transform and slant transform.

UNIT-II:

Image Enhancement: Spatial domain methods: Histogram processing, Fundamentals of Spatial filtering, Smoothing spatial filters, Sharpening spatial filters.

Frequency Domain Methods: Basics of filtering in frequency domain, Image smoothing, Image sharpening, Selective filtering.

UNIT-III:

Image Segmentation: Segmentation concepts, Point, Line and Edge Detection, Edge Linking using Hough Transform, Thresholding, Region Based segmentation.

Wavelet Based Image Processing: Introduction to wavelet Transform, Continuous wavelet Transform, Discrete wavelet Transform, Filter banks, Wavelet based image segmentation.

UNIT-IV:

Image Compression: Image compression fundamentals - Coding Redundancy, Spatial and Temporal redundancy, Compression models – Lossy and Lossless, Huffman coding, Arithmetic coding, LZW coding, Run length coding, Bit plane coding, Transform coding, Predictive coding, JPEG Standard.

UNIT-V:

Morphological Image Processing: Dilation and Erosion, Opening and closing, The Hit or Miss Transformation, Morphological algorithms.

Representation and Description: Boundary following, chain codes, polygonal approximation using minimum-perimeter polygons, boundary segments, skeleton, simple boundary descriptors, shape number, simple regional descriptors.

TEXT BOOKS:

1. Digital Image Processing, Rafael C. Gonzalez and Richard E. Woods, 4th Edition, Pearson, 2018
2. Digital Image Processing, S. Jayaraman, S. Esakkirajan, T. Veerakumar, 5th Edition, Tata McGraw-Hill, 2015

REFERENCES:

1. Digital Image Processing, William K. Pratt, 3rd Edition, John Wiley, 2007
2. Fundamentals of Digital Image Processing, A. K. Jain, 3rd Edition, PHI, 1989
3. Digital Image Processing using MATLAB, Rafael C. Gonzalez, Richard E. Woods and Steven L. Edding, 2nd Edition, Tata McGraw-Hill, 2010
4. Digital Image Processing and Computer Vision, Sonka, Hlavac, Boyle, Cengage Learning, 2008
5. Introduction to Image Processing and Analysis, John C. Russ, J. Christian Russ, CRC Press, 2008

B.Tech. VIII Semester

(22PE1EC420) CPLD AND FPGA ARCHITECTURES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Switching Theory and Logic Design

COURSE OBJECTIVES:

- To introduce digital design concepts through various Programmable Logic Devices
- To understand the CPLD and FPGA architectures in detail
- To analyze the physical design cycle in FPGA
- To know the various applications of CPLD and FPGAs

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Design digital applications using PLDs

CO-2: Analyze the architectural features of CPLDs, FPGAs

CO-3: Analyze Physical Design cycle for FPGA

CO-4: Implementation of various applications using FPGA

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	1	1	-	1	-	1	-	-	2	2	2
CO-2	3	2	3	1	1	1	-	1	-	1	-	-	2	2	2
CO-3	3	3	3	1	1	1	-	1	-	1	-	-	2	2	2
CO-4	3	2	3	1	1	1	-	1	-	1	-	-	2	2	2

UNIT-I:

Introduction to Programmable Logic Devices: Programmable logic devices (PLD), Simple Programmable Logic Devices (SPLD) – Read Only Memories, Programmable Logic Arrays (PLA), Programmable Array Logic (PAL): Registered PALs, Configurable PALS, Digital design using PLDs.

UNIT-II:

Complex Programmable Logic Devices: Features and applications of complex programmable logic devices, Altera Max - 7000 series and Altera FLEX logic- 10K series CPLD, Xilinx Cool Runner - II CPLD, CPLD Implementation of a parallel adder with accumulation.

UNIT-III:

Field Programmable Gate Arrays: Features and applications of FPGAs, advantages and disadvantages of FPGA, architecture of FPGA, technology trends, programming technologies, commercially Available FPGAs.

UNIT-IV:

SRAM Field Programmable Gate Arrays: SRAM Programming Technology, SRAM Programmable FPGAs: Xilinx XC4000, Spartan-3 FPGA Architectures.

Anti-Fuse Programmed FPGAs: Anti-fuse Programming technology, The Actel ACT1, ACT2 and ACT3 architectures.

UNIT-V:

Physical Design Implementation on FPGAs: FPGA Design flow, Physical Design cycle for FPGAs, Partitioning, Routing-non-segmented, segmented and staggered models. Design Applications: General design issues, Counter design Adder design using FPGA.

TEXT BOOKS:

1. Fundamentals of Logic Design, Charles H. Roth Jr, 5th Edition, Cengage Learning, 2004
2. Field Programmable Gate Array Technology, Stephen M. Trimberger, Springer International Edition, 1994

REFERENCES:

1. Algorithms for VLSI Physical Design Automation, Naveed Sherwani, 3rd Edition, Springer International Edition, 2005
2. Field-Programmable Gate Arrays, Stephen D. Brown, Springer, 1992

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EI410) AUTOMOTIVE INSTRUMENTATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To impart basic knowledge about classic and contemporary automobile electronics and instrumentation systems
- To impart the basic knowledge of the different subsystems of automotive
- To expose the control system applications to various modular operations of the automobiles

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the components of the automotive system and its associated electronics

CO-2: Appreciate the sensors, actuators, electronic engine control and measurement schemes used in the automotive subsystem

CO-3: Interpret the digital power train Vs conventional power train control systems

CO-4: Combine scientific principles and engineering concepts to formulate the control system required for different submodules of automotive systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	1	1	-	1	2	2	-	1	3	1	2	2
CO-2	3	2	1	2	2	-	1	2	2	-	1	3	3	2	2
CO-3	2	2	3	3	3	-	3	2	3	-	3	1	2	3	3
CO-4	3	3	3	3	2	-	3	2	3	-	3	1	2	3	3

UNIT-I:

Microcomputer Instrumentation and Control: Introduction to control, Microcomputer Fundamentals, Microcomputer Tasks & Operation, CPU Registers, Reading Instructions, Microcomputer Hardware, Microcomputer Applications in Automotive Systems, Instrumentation Applications of Microcomputers, Microcomputers in Control Systems, CAN Protocol

UNIT-II:

The Basics of Electronic Engine Control: Concept of an Electronic Engine Control System, Definition of Engine Performance Terms, Exhaust Catalytic Converters, Electronic Fuel-Control System, Analysis of Intake Manifold Pressure, Idle Speed Control, Electronic Ignition

UNIT-III:

Automotive Sensors and Actuators: Automotive Control System Applications of Sensors and Actuators, Throttle Angle Sensors, Temperature Sensors, Sensors for Feedback Control, Typical Coolant Sensors, Electric Motor Actuators, Ignition System

UNIT-IV:

Digital Powertrain Control Systems: Digital Engine Control, Digital Engine Control Features, Control Modes for Fuel Control, Electronic Ignition Control, Integrated Engine Control System, Hard Acceleration, Deceleration and Idle, Automatic Transmission Control, Torque Converter Lock-Up Control, Differential and Traction Control, Hybrid Electric Vehicle Powertrain Control.

UNIT-V:

Vehicle-Motion Controls: Representative Cruise Control System, Cruise Control Electronics, Antilock Braking System, Electronic Suspension System, Electronic Steering Control, Four-Wheel Steering

TEXT BOOK:

1. Understanding Automotive Electronics, William B. Ribbens, 7th Edition, Butterworth Heinemann, 2012

REFERENCES:

1. Automotive Electrical Equipment, Young A. P., Griffiths L., ELBS & New Press, 2010
2. Automotive Computers and Control System, Tom Weather Jr., Cland C. Hunter, Prentice Hall Inc., 2009
3. Automobile Electrical Equipment, Crouse W. H., McGraw-Hill, 2005
4. Understanding Automotive Electronic, Bechtold, SAE, 2010
5. Automotive Handbook, 9th Edition, Bentley Publishers, BOSCH, Germany, 2014